OSINACHI CHUKWUJAMA

Saint-Étienne, France
osinachi.chukwujama@gmail.com ⋄ +33 6 51 41 94 14
github.com/vicradon ⋄ linkedin.com/in/chukwujama-osinachi

PROFILE

Photonics master's student with a First Class degree in Mechanical Engineering and a strong mathematical foundation. I study machine learning from first principles: I have derived and implemented diffusion and flow-matching models in PyTorch, deployed vision models on embedded GPUs, and built physics simulations for optics. I want to spend a summer doing rigorous research in machine learning and data analysis alongside researchers at the Max Planck Institutes.

RESEARCH INTERESTS

Generative models (diffusion, flow matching, score-based models), probabilistic machine learning, computational imaging, world models.

EDUCATION

Erasmus Mundus Joint Master, Intelligent Photonics (iPSRS) September 2026 – 2028
Université Jean Monnet (France), University of Eastern Finland, Université Paris-Est Créteil
Relevant coursework: Fourier optics, digital image processing, probability and signal analysis, scientific computing with Python.

Bachelor of Engineering, Mechanical Engineering November 2017 – August 2023
Federal University of Technology, Owerri (FUTO), Nigeria. CGPA: 4.56/5.00 (First Class),
96th percentile in the department; best graduating student in Industrial and Production Engineering.

RESEARCH AND LABORATORY EXPERIENCE

Generative Modelling from First Principles 2026
Self-directed research

- Derived and implemented DDPM (forward/reverse process, ELBO, reparameterisation) and OT-conditional flow matching in PyTorch, covering training and sampling end to end.
- Studied the links between score matching (denoising and implicit variants, NCSN), flow matching and normalising flows; wrote a forward-diffusion noising tool in Go for synthetic data generation.
- Proposed a comparison of OT-CFM and DDPM on photonics/laser data (TPU Research Cloud application).

Spatial Light Modulator (SLM) Laboratory Practical 2026
iPSRS M1 coursework *Université Jean Monnet*

- Generated diffraction gratings and Fresnel lens phase patterns in MATLAB and displayed them on an SLM.
- Observed and analysed the resulting diffraction patterns, and computed the gratings' spatial frequencies from measurements.

Surveillance Drone with Real-Time Object Detection August 2023 – March 2024
Undergraduate final project and follow-up, supervised by Prof. O.O. Obiukwu *MEE, FUTO*

- Collected and annotated a three-class dataset, fine-tuned YOLOv5 (70% mAP) and ran live inference on a drone's camera stream via ONNX.
- Deployed the model on an NVIDIA Jetson Nano with TensorRT, cutting processing time by 44%.

Optics Playground In progress
Personal project, pre-zeemax.vercel.app

- Interactive 2D ray-tracing simulator with finite-size lenses, being extended into a Fourier optics bench.

SKILLS

Machine learning	PyTorch: DDPM, flow matching, score matching, CNN/ResNet training, transfer learning; ONNX, TensorRT
Mathematics	Probability, variational inference (ELBO), optimal transport for flow matching, Fourier analysis
Programming	Python (proficient), MATLAB, Go, TypeScript/JavaScript; Git, Linux, CI/CD, automated testing
Languages	English: C1 (IELTS 8.0)

WORK EXPERIENCE

Dstrct Group March 2025 – May 2026
Backend Engineer / Technical Lead *Lagos, Nigeria*

· Technical lead on an enterprise platform with a typed backend of nearly 1,000 files; worked directly with clients to turn requirements into engineering tasks.

Hotels.ng June 2024 – January 2026
Software/DevOps Engineer (part-time) *Lagos, Nigeria*

· Built a concurrent Python/Celery application distributing AI workloads over RabbitMQ; wrote CI/CD and monitoring tooling.

PUBLICATIONS

Quantum Computing in Artificial Intelligence: A Review of Quantum Machine Learning Algorithms, Path of Science. doi:10.22178/pos.117-25

Secure and Resilient Industrial IoT Architectures for Smart Manufacturing: A Comprehensive Review, JERR (2025). doi:10.9734/jerr/2025/v27i61548

Clinical Applications of 3D-Printed Medical Implants, JOPIR. doi:10.69613/22dn3m33

HONOURS, TEACHING AND COMMUNICATION

PTDF Scholarship (2018) for academic excellence.

Microsoft Certified Trainer and former **Microsoft Learn Student Ambassador**: trained over 300 students in software engineering and cloud technologies.

Director of Research, Mechanical Engineering, FUTO: ran hands-on training in machine learning, microcontrollers and 3D printing for 50+ students and staff. Co-founder of Hack4FUTO.

Technical writer at vicradon.hashnode.dev.